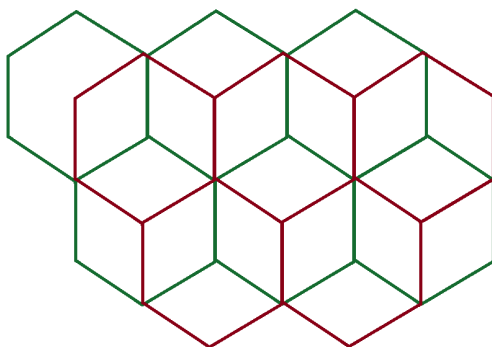


$$\phi = 360^\circ/3 = 120^\circ$$

$$\theta = \phi - 90^\circ = 120^\circ - 90^\circ = 30^\circ$$

$$\sin(30^\circ) = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{1/2l}{l} = 1/2$$

$$\text{adjacent}^2 + \text{opposite}^2 = \text{hypotenuse}^2$$

$$\text{adjacent}^2 + (l/2)^2 = (l)^2$$

$$\text{adjacent} = \frac{\sqrt{3}}{2}l$$

$$\cos(30^\circ) = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{\frac{\sqrt{3}}{2}l}{l} = \frac{\sqrt{3}}{2}$$